Supplementary Material

From Operator Rationales to Dispatch Policy: LLM-Enabled Semantic Reward Learning for V2G Scheduling

Manuscript ID: TSG-02158-2026 (revised manuscript)

This document provides the supporting material referenced in the manuscript and in the response letter.

Contents

- S1. PPO observation and action vectors
- S2. LLM annotation prompt
- S3. Label-validation checks of the annotation pipeline
- S4. Table III with standard deviations across seeds
- S5. Reward-model results on the common day split
- S6. Language-interface study: the 52 operator texts and outcomes
- S7. Fig. S1: battery-degradation retraining
- S8. Table S1: daily network-loss energy
- S9. Experimental settings

S1. PPO observation and action vectors

Source: `environments.py` (`state_dim = 33 + 6 + EV_type*2`, `action_dim = EV_type*2`, EV_type = 10), `DSO_environment/dso_flow.py` (network metrics block). The observation contains **no explicit time-of-day index**.

Policy observation s_t (59 dims)

index	dims	content	scaling
0–32	33	bus voltage magnitudes (p.u.), IEEE-33	none
33–34	2	upstream active / reactive power (kW, kvar)	/ 10^5 kW, / 10^5 kvar
35	1	available PV power (MW)	/ 10 MW
36	1	PV curtailment (MW)	/ 10 MW
37	1	active load curtailment (MW)	/ 10 MW
38	1	reactive load curtailment (Mvar)	/ 10 Mvar
39–48	10	current SOC of the 10 EV aggregations	[0, 1]
49–58	10	target SOC of the 10 EV aggregations	[0, 1]

In the manuscript, y_t denotes only the six non-voltage network entries (indices 33–38), not the full observation. Current and target SOCs are stored energy divided by the corresponding aggregate battery capacity. The two substation quantities are divided by 10^5 kW and 10^5 kvar; the other four power quantities are divided by 10 MW or 10 Mvar.

The hour t, operating period p_t, operator profile z_t, and reward-model context o_t are not explicit policy inputs. The policy is $\pi_\theta(a_t \mid s_t)$; the reward model is $\hat{r}_\phi(s_t, a_t, o_t)$. For fixed policy parameters, identical observations produce the same action distribution. Preference changes affect the policy through reward-driven parameter updates. Operating quantities can still correlate with time under the fixed scenario schedule.

Action a_t (20 dims, each in $[-1, 1]$)

index	dims	content
0–9	10	active charging (+) / discharging (−) fraction per EVCS, scaled by $P_{C/D}^{\max} = 22\,\text{kW} \times N_{EV}$

index	dims	content
10–19	10	reactive power fraction per EVCS, bounded by $\sqrt{S_{\max}^2 - P^2}$

Base fleet: $N_{EV} = 100$ per EVCS, $B_{EV} = 50\,\text{kWh} \times N_{EV}$, $S_{\max} = 40\,\text{kVA} \times N_{EV}$; EV-penetration experiments rescale N_{EV} (see `environments.py::set_ev_penetration`).

69-bus system

The 69-bus observation has the same structure: 69 bus-voltage magnitudes (p.u.), the six normalized network features above, and the current and target SOCs of the 10 EVCSs, giving $69 + 6 + 10 + 10 = 95$ dimensions.

S2. LLM annotation prompt

The complete annotation prompt template; its header lists the annotator settings.

```
---
annotator: qwen-turbo
temperature: 0.3
response_format: json_object
request_timeout_s: 180
max_retries: 10
batch: 2 episodes x 24 steps (all steps of each labeled day)
contrastive_pairs: 4
---

You are annotating EV charging / distribution-grid scheduling behavior for reward-model
↪  training.

# Batch Calibrated Preference Annotation Rubric

Use one shared scoring scale across the whole batch before assigning scalar labels.

- +7 to +10: clearly excellent. Target SOC is reached, service quality is good, voltage is
↪  safe, or the step clearly repairs an earlier severe service problem.
- +3 to +6: good. SOC gap shrinks materially, unmet energy decreases, near-departure EVs
↪  receive timely charging, or voltage violation is mitigated.
- -1 to +2: neutral or mild tradeoff. The state is stable, the issue is mild, or
↪  cost/service/voltage tradeoffs are acceptable.
- -3 to -5: moderate problem. SOC is not yet attained, voltage is biased, or cost-service
↪  tradeoff is poor, but there is still recovery room.
- -6 to -8: severe problem. Service is clearly insufficient, unmet SOC is high, time to
↪  departure is short, or voltage violation is clear.
- -9 to -10: extreme failure. Reserve for the worst small minority, such as near-departure
↪  severe undercharge combined with severe voltage violation or persistent service failure.

Important constraints:
1. Do not put every risky state into -8 or -9.
2. -9 to -10 can only be assigned to the most severe small minority inside a batch.
3. If a step improves SOC gap, unmet energy, or voltage violation relative to nearby states, it
↪  must score higher than a comparable worsening step, even if it has not fully recovered.
4. A non-catastrophic batch should contain positive, neutral, moderate-negative, and
↪  severe-negative labels when such examples exist.
5. If every step in a batch is negative, explain why there is no service recovery or acceptable
↪  tradeoff.
```

```
6. Neutral baseline states are not excellent labels. If target_reached=0,
↪  protected_good_anchor=0, soc_gap_shrink=0, unmet_decrease=0, voltage_mitigation=0, and
↪  expert_direction=neutral, then overall_score should not be excellent.
7. Low operating cost or safe voltage can move a neutral baseline slightly positive, but cannot
↪  move it to an excellent score without service-recovery, target-reached, voltage-mitigation,
↪  or protected-anchor evidence.

Hard anchor rules:
1. If target_reached=1 and there is no severe voltage problem, overall_score must be >= +3.
2. If soc_gap_shrink=1 and voltage did not worsen, overall_score must be >= 0.
3. If unmet_decrease=1 or a fast-departure vehicle receives timely charging, overall_score
↪  should be >= 0.
4. Severe near-departure shortfall should be <= -6.
5. Severe voltage violation should be <= -5 unless voltage_mitigation=1 and the step is clearly
↪  repairing voltage.
6. The batch input includes protected_good_anchor. If protected_good_anchor=1, overall_score
↪  must be >= 0.
7. If a recovery flag is present but protected_good_anchor=0, it is an exception case such as
↪  severe near-departure shortfall, severe voltage, or extreme cost. It may be negative only
↪  if label_reason names the exception.
8. Treat voltage_violation_severity < 0.09 as not severe. It may still reduce voltage_score,
↪  but by itself it is not enough to make a protected_good_anchor negative.
9. Treat voltage_violation_severity >= 0.10 as severe and >= 0.13 as extreme.
10. If voltage_mitigation=1 and voltage_violation_severity < 0.09, overall_score must be >= 0
↪  unless operating_cost is extreme.
11. Do not rely on the input rank_group alone. The explicit anchor and exception flags override
↪  the rank_group when deciding whether a step may be negative.

Score relationship:
- service_score measures EV charging service quality.
- voltage_score measures grid safety.
- cost_score measures operating-cost reasonableness.
- overall_score is service-dominant, but it must still account for voltage and cost.
- A high service_score does not automatically imply a high overall_score if voltage is severely
↪  unsafe or cost is extremely unreasonable.

Cost discipline (v6 addendum, used only when key_steps carry
cost_ratio_vs_reference, cost_p50_reference, cost_p90_reference,
delta_operating_cost, and cost_spike_unjustified):

12. cost_ratio_vs_reference = operating_cost / cost_p50_reference (the
    scenario-period median cost from a reference policy). 1.0 is the
    reference. > 2.0 is high. > 3.0 is extreme. Use this absolute ratio
    as the cost trade-off signal, NOT just the intra-batch cost_q90.
13. If cost_ratio_vs_reference > 2.0 and the step has no voltage_mitigation,
    no target_reached, and no protected_good_anchor, then overall_score
    must not be greater than +1, even if soc_gap_shrink=1 or
    unmet_decrease=1. Service progress paid for by expensive grid imports
    is not a positive label.
14. If cost_ratio_vs_reference > 3.0 and there is no recovery flag
    (target_reached / voltage_mitigation / protected_good_anchor), then
    overall_score must be <= -3.
15. cost_spike_unjustified = 1 marks a step whose cost is at least 2x the
    scenario reference AND has no voltage_mitigation AND no
    terminal_shortfall_risk AND no target_reached. Such a step must not
    score above 0.
```

```
16. A cost ramp (delta_operating_cost > 0.5 * cost_p90_reference) is
    evidence to reduce cost_score even when overall_score remains positive
    due to service. cost_score in this case should be at most 1.
17. Apply cost-discipline clauses 13-16 AFTER applying clauses 1-7. They
    can move a "+5 service-recovery" step down to +1, but they cannot
    cancel target_reached=1 below the +3 floor unless voltage is also
    severely unsafe.


Expert-anchored v5 fields:
Each key step may include expert_direction, expert_reasons, expert_confidence,
↪ expert_red_lines, expert_soft_guidance, and the compatibility fields expert_evaluation,
↪ expert_score_hint, and expert_priority.
These fields are expert calibration guidance derived from the step metrics and hard anchors,
↪ but they are not exact answer keys. Treat them as direction and red-line guidance. The LLM
↪ still decides the final scalar label using step metrics, pairwise context, and the expert
↪ guidance together.

When expert fields are present:
- Every key step must have an expert evaluation.
- Use expert_direction as the main directional prior: positive, negative, neutral, or mixed.
- Use expert_reasons to identify the major service, voltage, cost, and urgency evidence.
- Use expert_confidence to decide how strict the expert hint is. High-confidence red lines must
↪ be followed. Medium/low confidence mixed tradeoffs can be weighed by the LLM.
- Use expert_red_lines as non-negotiable hard constraints unless the step metrics show that the
↪ expert generator is clearly wrong.
- Use expert_soft_guidance as interpretation help, not as a precise score.
- The compatibility field expert_score_hint is soft guidance unless expert_confidence=high and
↪ a red line applies.
- If the final score differs from expert_direction, explain why in expert_alignment_reason.
- Add expert_alignment and expert_alignment_reason to every scalar_labels item.
- expert_alignment must be one of: followed, soft_exception, red_line_exception.
- Use followed when the label follows the expert direction or red line.
- Use soft_exception when the label departs from soft expert guidance with a reasonable metric
↪ or pairwise explanation.
- Use red_line_exception only when the final label intentionally violates a high-confidence red
↪ line because the step metrics prove the expert generator is wrong. This will usually fail
↪ hard audit.
- If a high-confidence red line forces you to cap or lower a score and you obey that red line,
↪ use expert_alignment=followed, not red_line_exception.
- self_check must also include missing_expert_alignment_count,
↪ high_confidence_red_line_violation_count, expert_direction_mismatch_count,
↪ expert_contradiction_count, and expert_score_hint_violation_count.


Follow this order. Do not jump directly to scalar scores.

Step 1: Episode-level ranking
Rank the episodes from best to worst overall scheduling quality using SOC attainment, unmet
↪ SOC, time-to-departure risk, voltage violation, cost tradeoff, and service recovery.

Step 2: Step-group ranking
Within each episode, rank the provided key steps from best to worst service quality and grid
↪ safety. Mark each step as one of:
positive_anchor, neutral_tradeoff, moderate_failure, severe_failure.
```

```
Step 3: Pairwise comparison
For each contrastive pair, decide which step is better. Prefer the step with smaller SOC gap,
↪ longer time to departure, lower unmet SOC, lower voltage severity, lower service deficit,
↪ or clearer service recovery.

Step 4: Scalar labels
Only after ranking and pairwise comparison, output scores. If A is judged clearly better than
↪ B, A's overall_score must be higher than B's.

Output only JSON with keys:
- episode_level_ranking
- step_group_ranking
- pairwise_comparisons
- scalar_labels
- self_check

Each scalar_labels item must contain:
episode_id, step, service_score, voltage_score, cost_score, overall_score, label_reason,
↪ rank_group

If expert fields are present in key_steps, each scalar_labels item must additionally contain:
expert_alignment, expert_alignment_reason

The self_check object must contain:
expected_scalar_labels, actual_scalar_labels, good_anchor_negative_count,
↪ pairwise_violation_count, extreme_negative_count, positive_count, neutral_count,
↪ negative_count

If expert fields are present in key_steps, self_check must additionally contain:
missing_expert_alignment_count, high_confidence_red_line_violation_count,
↪ expert_direction_mismatch_count, expert_contradiction_count,
↪ expert_score_hint_violation_count

Before returning JSON, verify:
- actual_scalar_labels equals expected_scalar_labels.
- For every contrastive pair, better step overall_score is greater than worse step
↪ overall_score.
- Ties are pairwise violations. If a better step and worse step have equal scores, raise the
↪ better step or lower the worse step before returning.
- good_anchor_negative_count counts protected_good_anchor steps with negative overall_score,
↪ and it must be 0.
- Recovery steps with protected_good_anchor=0 may be negative only when label_reason names
↪ severe shortfall, severe voltage, or extreme cost.
- If expert_direction is neutral and there is no target_reached, soc_gap_shrink,
↪ unmet_decrease, voltage_mitigation, or protected_good_anchor evidence, do not assign an
↪ excellent score.
- If expert_confidence is high and expert_red_lines contains a rule, obey it and mark
↪ expert_alignment=followed. If you intentionally violate the red line because the metrics
↪ prove the expert generator is wrong, use expert_alignment=red_line_exception and explain
↪ the exact metric conflict.

Batch input:
{batch_json}
```

S3. Label-validation checks of the annotation pipeline

Source: `analysis/qwen_batch_calibrated_annotation.py`. The `--validation-mode` switch (`full` = default, `format_only`, `none`) is the ablation of Fig. 10(e) in the manuscript.

Format check (stage 1): format / completeness (per batch)

The response must parse as a JSON object with a `scalar_labels` array covering every key step of the batch; otherwise the batch is retried (`full`, `format_only`).

```
def parse_json_object(text: str) -> Dict[str, Any]:
    s = strip_json_fences(text)
    try:
        obj = json.loads(s)
        if isinstance(obj, dict):
            return obj
    except json.JSONDecodeError:
        pass

    start = s.find("{")
    end = s.rfind("}")
    if start >= 0 and end > start:
        obj = json.loads(s[start : end + 1])
        if isinstance(obj, dict):
            return obj
    raise ValueError("Could not parse a JSON object from LLM response")
```

```
def scalar_labels_from_answer(answer: Dict[str, Any]) -> List[Dict[str, Any]]:
    labels = answer.get("scalar_labels")
    if labels is None and "labels" in answer:
        labels = answer["labels"]
    if not isinstance(labels, list):
        raise ValueError("Missing scalar_labels list")
    return labels
```

Step-consistency check (stage 2): per-step consistency with the operator cue (`full` only triggers retries)

```
def score_hint_violated(score: float, hint: Any) -> bool:
    hint_s = str(hint or "").strip().lower()
    if hint_s == "positive_ge_3":
        return score < 3
    if hint_s == "non_negative_0_to_6":
        return score < 0
    if hint_s == "neutral_minus1_to_plus2":
        return score < -1 or score > 2
    if hint_s == "mixed_tradeoff_minus3_to_plus3":
        return score < -3 or score > 3
    if hint_s == "mixed_tradeoff_minus4_to_plus3":
        return score < -4 or score > 3
    if hint_s == "mitigated_near_departure_tradeoff_minus5_to_plus3":
        return score < -5 or score > 3
    if hint_s == "moderate_negative_minus3_to_minus5":
        return score < -5 or score > 2
    if hint_s == "severe_shortfall_le_minus6":
        return score > -6
    if hint_s == "severe_voltage_le_minus5":
        return score > -5
```

```
    if hint_s == "severe_negative_minus6_to_minus8":
        return score > -6 or score < -8
    if hint_s == "extreme_negative_le_minus8":
        return score > -8
    return False

def high_confidence_red_line_violated(score: float, step: Dict[str, Any]) -> bool:
    confidence = str(step.get("expert_confidence", "")).strip().lower()
    if confidence != "high":
        return False
    red_lines = str(step.get("expert_red_lines", "")).lower()
    if "protected_good_anchor must be non negative" in red_lines and score < 0:
        return True
    if "severe near departure shortfall" in red_lines and not step_has_clear_recovery(step) and
    ↪  score > -3:
        return True
    if "severe voltage violation" in red_lines and int(step.get("voltage_mitigation", 0)) != 1
    ↪  and score > 2:
        return True
    if "neutral baseline should not receive excellent score" in red_lines and not
    ↪  step_has_clear_recovery(step) and score >= 7:
        return True
    return False

def expert_direction_mismatched(score: float, step: Dict[str, Any]) -> bool:
    direction = str(step.get("expert_direction", "")).strip().lower()
    if direction == "positive":
        return score < 0
    if direction == "negative":
        return score > 2
    if direction == "neutral":
        return score >= 6 or score < -3
    if direction == "mixed":
        return False
    return False
```

Dataset-level check (stage 3): batch-level self-check and dataset-level audit

`compute_self_check` counts pairwise-order violations, negative scores on protected good anchors, missing expert-alignment fields and high-confidence red-line violations; any positive count triggers a retry in `full` mode. After annotation, `audit_labels` checks the label distribution (positive ≥ 5 %, neutral ≥ 3 %, extreme-negative ≤ 15 %), parsed-label coverage (≥ 90 % for partial finalize) and the soft violation rate (≤ 10 %).

```
def compute_self_check(example: Dict[str, Any], labels: List[Dict[str, Any]]) -> Dict[str,
↪  Any]:
    label_items = {
        (int(item["episode"]), int(item["step"])): item
        for item in labels
    }
    label_lookup = {key: float(item["overall_score"]) for key, item in label_items.items()}
    key_steps = example["batch_input"]["key_steps"]
    step_lookup = {
        (int(item["episode"]), int(item["step"])): item
        for item in key_steps
    }
    good_anchor_negative_count = 0
    for key, step in step_lookup.items():
        score = label_lookup.get(key)
```

```
    if score is None:
        continue
    if "protected_good_anchor" in step:
        is_good_anchor = int(step.get("protected_good_anchor", 0)) == 1
    else:
        is_good_anchor = any(
            int(step.get(name, 0)) == 1
            for name in ("target_reached", "soc_gap_shrink", "unmet_decrease",
            ↪  "voltage_mitigation")
        )
    if is_good_anchor and score < 0:
        good_anchor_negative_count += 1

pairwise_violation_count = 0
for pair in example["batch_input"].get("contrastive_pairs", []):
    better = (int(pair["better_episode"]), int(pair["better_step"]))
    worse = (int(pair["worse_episode"]), int(pair["worse_step"]))
    better_score = label_lookup.get(better)
    worse_score = label_lookup.get(worse)
    if better_score is None or worse_score is None or better_score <= worse_score:
        pairwise_violation_count += 1

scores = [float(item["overall_score"]) for item in labels]
computed = {
    "expected_scalar_labels": len(key_steps),
    "actual_scalar_labels": len(labels),
    "good_anchor_negative_count": int(good_anchor_negative_count),
    "pairwise_violation_count": int(pairwise_violation_count),
    "extreme_negative_count": int(sum(score <= -8 for score in scores)),
    "positive_count": int(sum(score > 0 for score in scores)),
    "neutral_count": int(sum(-1 <= score <= 2 for score in scores)),
    "negative_count": int(sum(score < 0 for score in scores)),
}
if example_has_expert_fields(example):
    missing_expert_alignment_count = 0
    expert_contradiction_count = 0
    expert_score_hint_violation_count = 0
    high_confidence_red_line_violation_count = 0
    expert_direction_mismatch_count = 0
    for step in key_steps:
        key = (int(step["episode"]), int(step["step"]))
        item = label_items.get(key)
        if item is None:
            missing_expert_alignment_count += 1
            expert_contradiction_count += 1
            high_confidence_red_line_violation_count += 1
            continue
        score = float(item["overall_score"])
        alignment = str(item.get("expert_alignment", "")).strip().lower()
        alignment_reason = str(item.get("expert_alignment_reason", ""))
        combined_reason = f"{item.get('label_reason', '')} {alignment_reason}"
        if not alignment:
            missing_expert_alignment_count += 1
        if high_confidence_red_line_violated(score, step):
            high_confidence_red_line_violation_count += 1
            expert_contradiction_count += 1
```

```
            if expert_direction_mismatched(score, step):
                expert_direction_mismatch_count += 1
            violated = score_hint_violated(score, step.get("expert_score_hint"))
            exception_named = severe_exception_named(combined_reason)
            if violated and not exception_named:
                expert_score_hint_violation_count += 1
            if alignment in {"red_line_exception"}:
                high_confidence_red_line_violation_count += 1
                expert_contradiction_count += 1
            if alignment in {"overridden", "exception", "partially_followed", "contradicted",
            ↪  "not_followed"} and not exception_named:
                expert_contradiction_count += 1
        computed.update(
            {
                "missing_expert_alignment_count": int(missing_expert_alignment_count),
                "high_confidence_red_line_violation_count":
                ↪  int(high_confidence_red_line_violation_count),
                "expert_direction_mismatch_count": int(expert_direction_mismatch_count),
                "expert_contradiction_count": int(expert_contradiction_count),
                "expert_score_hint_violation_count": int(expert_score_hint_violation_count),
            }
        )
    return computed
```

S4. Table III with standard deviations across seeds

Policies frozen after 10,000 episodes, evaluated deterministically on 24 held-out days; each entry is mean ± standard deviation across the three seeds (503, 509, 521). Columns 33-V to 69-C denote the 33-bus and 69-bus systems under the voltage-first (V), service-first (S), and cost-first (C) preferences.

Metric	Method	33-V	33-S	33-C	69-V	69-S	69-C
Return per day	Semantic Reward	-1.42 ± 0.03	-0.64 ± 0.04	0.49 ± 0.01	-1.24 ± 0.13	-0.76 ± 0.02	0.06 ± 0.04
	Expert Reference	-1.96 ± 0.02	-0.83 ± 0.11	0.48 ± 0.00	-1.60 ± 0.07	-0.98 ± 0.00	0.01 ± 0.00
	Score-only Reward	-2.81 ± 0.12	-1.05 ± 0.18	-2.40 ± 1.45	-4.46 ± 2.93	-1.27 ± 0.21	-3.98 ± 1.04
	Price–Service Adaptive Reward (PSA)	-3.27 ± 0.04	-0.96 ± 0.00	0.48 ± 0.00	-3.54 ± 0.07	-1.06 ± 0.03	0.01 ± 0.00
Purchase cost (kCNY/day)	Semantic Reward	46.2 ± 0.1	45.8 ± 0.2	44.8 ± 0.0	49.7 ± 0.2	48.4 ± 0.3	47.0 ± 0.1
	Expert Reference	47.7 ± 0.3	45.5 ± 0.6	44.8 ± 0.0	55.4 ± 0.5	49.7 ± 0.8	47.1 ± 0.0
	Score-only Reward	48.5 ± 1.5	48.0 ± 2.6	47.4 ± 1.1	53.8 ± 4.3	54.3 ± 3.4	51.7 ± 1.6
	Price–Service Adaptive Reward (PSA)	44.8 ± 0.0	44.8 ± 0.0	44.8 ± 0.0	47.1 ± 0.0	47.1 ± 0.0	47.1 ± 0.0

Metric	Method	33-V	33-S	33-C	69-V	69-S	69-C
Waiting deficit (SOC-h/day)	Semantic Reward	2.78 ± 0.04	2.63 ± 0.03	2.70 ± 0.12	2.95 ± 0.01	2.80 ± 0.08	2.87 ± 0.05
	Expert Reference	2.71 ± 0.12	2.18 ± 0.03	2.27 ± 0.05	2.75 ± 0.04	2.19 ± 0.01	2.41 ± 0.01
	Score-only Reward	2.63 ± 0.23	2.39 ± 0.15	4.08 ± 2.06	2.47 ± 0.08	2.86 ± 0.27	2.91 ± 0.17
	Price–Service Adaptive Reward (PSA)	2.36 ± 0.11	2.34 ± 0.11	2.25 ± 0.01	2.47 ± 0.03	2.48 ± 0.04	2.42 ± 0.01
Voltage violation rate	Semantic Reward	0.008 ± 0.004	0.019 ± 0.011	0.057 ± 0.009	0.007 ± 0.003	0.010 ± 0.001	0.096 ± 0.001
	Expert Reference	0.033 ± 0.005	0.049 ± 0.023	0.076 ± 0.001	0.010 ± 0.004	0.070 ± 0.008	0.111 ± 0.003
	Score-only Reward	0.031 ± 0.003	0.043 ± 0.014	0.413 ± 0.290	0.119 ± 0.100	0.072 ± 0.029	0.414 ± 0.010
	Price–Service Adaptive Reward (PSA)	0.071 ± 0.005	0.069 ± 0.005	0.075 ± 0.002	0.107 ± 0.003	0.103 ± 0.008	0.112 ± 0.002
Network loss (MWh/day)	Semantic Reward	6.6 ± 0.3	6.0 ± 0.3	4.3 ± 0.0	10.1 ± 0.4	7.7 ± 0.6	4.9 ± 0.2
	Expert Reference	9.7 ± 0.7	5.6 ± 1.1	4.3 ± 0.0	20.4 ± 0.9	10.4 ± 1.6	5.1 ± 0.0
	Score-only Reward	10.4 ± 2.3	9.2 ± 4.2	9.2 ± 2.5	17.3 ± 7.4	18.0 ± 5.5	13.6 ± 2.6
	Price–Service Adaptive Reward (PSA)	4.3 ± 0.0	4.3 ± 0.0	4.3 ± 0.0	5.1 ± 0.0	5.1 ± 0.1	5.1 ± 0.0
Service limits met (share of days)	Semantic Reward	0.94 ± 0.02	0.97 ± 0.02	0.90 ± 0.10	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
	Expert Reference	0.96 ± 0.04	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
	Score-only Reward	0.90 ± 0.13	1.00 ± 0.00	0.67 ± 0.58	0.94 ± 0.10	0.81 ± 0.17	0.96 ± 0.04
	Price–Service Adaptive Reward (PSA)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00

S5. Reward-model results on the common day split

Both reward models are fitted on the same days, early-stopped on the same 20 days, and evaluated on the same 20 test days; three seeds per system and preference (18 configurations). Step-level Pearson correlation, mean ± SD over the 18 configurations; the last column counts configurations in which the semantic model is higher.

Fitting days (steps)	Subset	Score-only	Semantic	Semantic higher
10 (240)	fitting	0.910 ± 0.135	0.994 ± 0.012	14/18
10 (240)	early-stopping	0.621 ± 0.099	0.855 ± 0.038	18/18
10 (240)	test	0.628 ± 0.075	0.856 ± 0.027	18/18
20 (480)	fitting	0.971 ± 0.025	0.992 ± 0.009	16/18
20 (480)	early-stopping	0.722 ± 0.057	0.905 ± 0.015	18/18
20 (480)	test	0.710 ± 0.056	0.895 ± 0.017	18/18
60 (1440)	fitting	0.937 ± 0.043	0.989 ± 0.010	17/18
60 (1440)	early-stopping	0.827 ± 0.048	0.936 ± 0.010	18/18
60 (1440)	test	0.809 ± 0.049	0.929 ± 0.012	18/18

Clear-pair day-ranking accuracy on the 20 test days (fraction of test-day pairs with clearly different labeled scores ordered correctly), mean over the 18 configurations:

Fitting days	Score-only	Semantic
10	0.661	0.813
20	0.679	0.831
60	0.788	0.878

S6. Language-interface study: the 52 operator texts and outcomes

The test set contains 52 simulated operator texts across nine expression types: 40 specify an unambiguous priority order among voltage security, charging service, and purchase cost, and 12 contain missing information or contradictory requirements. Expected output: the priority order, or a clarification request for incomplete or contradictory texts. The Writer column records the three independent LLM-driven text sources (A, B, C). Interfaces: keyword parser (v0), keyword parser with clarification fallback (v1), qwen-turbo (initial run) and qwen3.8-flash (repeat), both at temperature 0 with three requests per text; the LLM received the system prompt and the verbatim text only.

In the table, V, S, and C denote voltage security, charging service, and purchase cost; for example, V > S > C ranks voltage first, service second, and cost last.

id	Type	Writer	Expected	Parser v0	Parser v1	qwen-turbo	qwen3.8-flash
r01	Negation or concession	A	V > S > C	invalid	clarify	V > S > C	V > S > C
r02	Multi-party ruling	C	C > S > V	invalid	clarify	C > S > V	C > S > V
r03	Cross-sentence reference	A	C > S > V	invalid	clarify	C > S > V	C > S > V
r04	Contradiction (should clarify)	C	clarify	V > S > C	clarify	clarify	clarify
r05	Cross-sentence reference	C	C > S > V	clarify	clarify	C > S > V	C > S > V
r06	Standard wording	A	V > S > C	V > S > C	V > S > C	V > S > C	V > S > C
r07	Standard wording	A	V > S > C	clarify	clarify	V > S > C	V > S > C
r08	Superseded instruction	C	V > S > C	clarify	clarify	V > S > C	V > S > C
r09	Standard wording	A	C > S > V	C > S > V	C > S > V	C > S > V	C > S > V
r10	Multi-paragraph handover	B	V > S > C	clarify	V > S > C	V > S > C	V > S > C

id	Type	Writer	Expected	Parser v0	Parser v1	qwen-turbo	qwen3.8-flash
r11	Background + final directive	A	C > S > V	C > S > V	C > S > V	C > S > V	C > S > V
r12	Superseded instruction	A	V > S > C	invalid	V > S > C	V > S > C	V > S > C
r13	Contradiction (should clarify)	A	clarify	clarify	clarify	clarify	clarify
r14	Cross-sentence reference	B	C > S > V	clarify	clarify	C > S > V	C > S > V
r15	Missing priority (should clarify)	A	clarify	invalid	clarify	clarify	clarify
r16	Negation or concession	B	V > S > C	V > S > C	clarify	V > S > C	V > S > C
r17	Multi-party ruling	A	C > S > V	clarify	clarify	C > S > V	C > S > V
r18	Superseded instruction	B	V > S > C	V > C > S	V > S > C	V > S > C	V > S > C
r19	Negation or concession	B	C > S > V	invalid	C > S > V	C > S > V	C > S > V
r20	Multi-paragraph handover	A	V > S > C	invalid	V > S > C	V > S > C	V > S > C
r21	Cross-sentence reference	C	V > S > C	clarify	clarify	V > C > S	V > S > C
r22	Superseded instruction	A	C > S > V	clarify	clarify	C > S > V	C > S > V
r23	Background + final directive	B	V > S > C	V > S > C	V > S > C	V > S > C	V > S > C
r24	Negation or concession	A	C > S > V	invalid	clarify	C > S > V	C > S > V
r25	Multi-paragraph handover	B	C > S > V	C > S > V	clarify	C > S > V	C > S > V
r26	Background + final directive	B	C > S > V	clarify	clarify	C > S > V	C > S > V
r27	Multi-paragraph handover	C	V > S > C	clarify	V > S > C	V > S > C	V > S > C
r28	Negation or concession	C	V > S > C	invalid	clarify	V > S > C	V > S > C
r29	Cross-sentence reference	A	V > S > C	clarify	clarify	V > S > C	V > S > C
r30	Multi-party ruling	B	C > S > V	invalid	clarify	C > S > V	C > S > V
r31	Multi-paragraph handover	C	C > S > V	C > V > S	C > S > V	C > S > V	C > S > V
r32	Missing priority (should clarify)	C	clarify	clarify	clarify	S > V > C	clarify
r33	Missing priority (should clarify)	B	clarify	invalid	clarify	V > S > C	clarify
r34	Superseded instruction	C	C > S > V	invalid	clarify	C > S > V	C > S > V
r35	Missing priority (should clarify)	C	clarify	clarify	clarify	V > S > C	clarify
r36	Cross-sentence reference	B	V > S > C	invalid	clarify	V > S > C	V > S > C
r37	Superseded instruction	B	C > S > V	clarify	C > S > V	C > S > V	C > S > V
r38	Multi-paragraph handover	A	C > S > V	invalid	clarify	C > S > V	C > S > V
r39	Multi-party ruling	A	V > S > C	invalid	clarify	V > S > C	V > S > C
r40	Multi-party ruling	C	V > S > C	V > S > C	clarify	V > S > C	V > S > C
r41	Missing priority (should clarify)	A	clarify	clarify	clarify	clarify	clarify
r42	Background + final directive	A	V > S > C	V > S > C	clarify	V > S > C	V > S > C
r43	Contradiction (should clarify)	A	clarify	invalid	clarify	clarify	clarify

id	Type	Writer	Expected	Parser v0	Parser v1	qwen-turbo	qwen3.8-flash
r44	Contradiction (should clarify)	C	clarify	V > C > S	clarify	clarify	clarify
r45	Contradiction (should clarify)	B	clarify	clarify	clarify	V > C > S	clarify
r46	Contradiction (should clarify)	B	clarify	clarify	clarify	V > S > C	clarify
r47	Multi-party ruling	B	V > S > C	invalid	clarify	V > S > C	V > S > C
r48	Negation or concession	C	C > S > V	clarify	C > S > V	C > S > V	C > S > V
r49	Background + final directive	C	C > S > V	clarify	C > S > V	C > S > V	C > S > V
r50	Missing priority (should clarify)	B	clarify	invalid	clarify	S > V > C	clarify
r51	Standard wording	A	C > S > V	C > S > V	C > S > V	C > S > V	C > S > V
r52	Background + final directive	C	V > S > C	invalid	V > S > C	V > S > C	V > S > C

Texts

r01 (Negation or concession, writer A, expected: voltage > service > cost)

I want to be clear about what this evaluation is not. It is not a cost-minimisation exercise, and it is not mainly about pleasing drivers either. Voltage comes before anything else: a dispatch that keeps every bus comfortably within its limits beats one that does not, even if the safer one leaves some vehicles less well served. Paying somewhat more for electricity is acceptable if that is what it takes to give drivers better service, shorter waits at low charge and fuller charge at departure, because service ranks above the purchase bill. Do not read this as permission to ignore cost; it still matters, just less than the other two. And do not let service gains justify pushing voltages toward the edge, because that trade goes the other way.

r02 (Multi-party ruling, writer C, expected: cost > service > voltage)

Three views came in on the evaluation criteria. The grid engineer asked that keeping bus voltages within limits be the main test of a good dispatch day. The fleet coordinator would prefer that vehicles be served first, with minimal waiting at low charge and no one leaving short of what they requested. The finance lead asked that electricity purchase cost be the headline measure. I have the final say on this, and after weighing all three, I am siding with finance: cost is the leading criterion. Of the two remaining, the fleet coordinator's service concern ranks above the voltage question. Voltage stays in the evaluation as the final consideration, so the grid engineer's request is not discarded, only placed behind the others. That ruling holds for the whole day.

r03 (Cross-sentence reference, writer A, expected: cost > service > voltage)

Let me lay out the three concerns before ranking them. One is keeping bus voltages within their limits. Another is driver service at the chargers, covering both time spent waiting at low charge and departing short of the requested energy. The last is what we pay for electricity bought from upstream. For today's whole-day evaluation, the last of these is the one that matters most. The second one I mentioned follows it. The one I listed first is still part of the judgment, but it ranks below the other two. Put simply, the order I want is the reverse of the order in which I introduced them.

r04 (Contradiction (should clarify), writer C, expected: clarify)

Here is the evaluation guidance for the day. Voltage security is the top priority: keeping every bus within its limits matters more than anything else in the assessment. Service to vehicles comes next, with short waits at low charge and departures with the requested energy. Electricity purchase cost is the least important of the three. At the same time, and this is equally firm, the purchase cost of electricity must be the leading criterion today, ahead of both service and voltage, because the budget review depends on it. Both of these instructions come from me and both stand; please apply them together across the whole day.

r05 (Cross-sentence reference, writer C, expected: cost > service > voltage)

When reviewing a dispatch day, you'll be weighing voltage security on the feeder against the quality of charging service, and both against what the electricity cost us. Start from the last of those. It is the one I care about most, and a day that buys energy expensively should not score well just because it looks good on other fronts. As for the first two, the latter takes precedence over the former: vehicles waiting long at low charge, or departing short of their requested energy, is a bigger concern than voltages drifting toward their limits. That is not to say the former can be ignored; it remains part of the judgement, simply at the bottom of it.

r06 (Standard wording, writer A, expected: voltage > service > cost)

Voltage security comes first, EV charging service second, and electricity purchase cost last.

r07 (Standard wording, writer A, expected: voltage > service > cost)

Prioritise keeping bus voltages within limits, then EV charging service, then electricity purchase cost.

r08 (Superseded instruction, writer C, expected: voltage > service > cost)

Please disregard the evaluation guidance circulated earlier this month. That note asked for dispatch to be judged mainly on how cheaply the energy was bought, with driver service second and the grid last. It has been withdrawn in full after the review meeting and should not be used for anything going forward. The arrangement that replaces it is as follows. Across the whole day, voltage security comes first: a dispatch that keeps every bus within its voltage limits is better than one that does not, before anything else is considered. Charging service for the vehicles is the next consideration, covering both how long cars sit waiting while their batteries are low and whether they depart with the charge they requested. Electricity purchase cost is the least important consideration, though it is not to be ignored.

r09 (Standard wording, writer A, expected: cost > service > voltage)

Minimise electricity purchase cost above all, then serve EV charging needs, and treat voltage security as the lowest priority.

r10 (Multi-paragraph handover, writer B, expected: voltage > service > cost)

Shift handover, night to day.

Overnight was quiet. Two chargers at the north hub faulted and were reset; both are back in service. No customer complaints were logged.

Yesterday afternoon we were told to treat cost as the top priority while the price review was running. That review finished last night and the control room supervisor cancelled the instruction, so it does not carry over.

The coming day is evaluated under the supervisor's standing guidance for this feeder. The day is judged first on voltage security, keeping all bus voltages within limits. Charging service is the next consideration: avoid long waits for vehicles at low charge and make sure each car leaves with its requested energy. Electricity purchase cost is the last of the three.

The forecast is cloudy, so expect little help from rooftop solar.

r11 (Background + final directive, writer A, expected: cost > service > voltage)

For the record, the utility's wholesale contract was renegotiated this quarter, and our upstream purchase price has become the line item management reviews every week. The feeder itself has been comparatively healthy lately; voltage complaints have been rare and the engineers describe the margins as comfortable. Station attendants report that charging customers are mostly satisfied, though some still mention long stretches at low battery. With that in mind, my instruction for evaluating the whole day is this: what we pay for electricity from upstream is the primary concern. Charging service for the vehicles, meaning not leaving them waiting while nearly empty and not sending them off without the energy they requested, ranks next. Voltage security still counts, but it sits at the bottom of the list for today's judgment.

r12 (Superseded instruction, writer A, expected: voltage > service > cost)

Please disregard the guidance I circulated yesterday. In that note I said purchase cost should lead and that voltage could be treated as the last consideration; that arrangement is withdrawn and should not be applied to anything you evaluate from now on. After talking it over with the network planning group, the current arrangement is different. Voltage security on the feeder takes precedence over everything else in this assessment. Charging service for EV owners, that is, how long they sit with a low battery and whether they depart with the charge they asked for, is the second consideration. Electricity purchase cost comes after both of those. Judge the full day's dispatch on that basis only, and treat the earlier note as void.

r13 (Contradiction (should clarify), writer A, expected: clarify)

Handover for the evaluation desk. To be clear about the current ordering of our three concerns: charging service for the vehicles matters more than electricity purchase cost, and purchase cost in turn matters more than keeping bus voltages within limits. Voltage security, meanwhile, matters more than charging service. All three of these comparisons apply at the same time to the whole day, and none of them is a past arrangement that has been withdrawn. Please judge today's dispatch so that every one of these statements is respected exactly as written, without dropping any of them.

r14 (Cross-sentence reference, writer B, expected: cost > service > voltage)

Consider the three things this day will be judged on: what we pay for electricity, the charging service drivers receive, and whether bus voltages stay within limits. Of these, the one listed first is tomorrow's main criterion. The second item, which covers both how long a car waits at low charge and whether it leaves with the energy requested, comes right after it. The last item on that list is the least important of the three. It should not be ignored, but where it competes with either of the others, it gives way. In short, the item I mentioned first beats the one I mentioned second, and that one in turn beats the third.

r15 (Missing priority (should clarify), writer A, expected: clarify)

For today's evaluation, voltage security on the feeder clearly comes first; nobody disputes that. What I have not settled is how charging service and electricity purchase cost compare with each other. Finance and customer relations each make a fair case, one about the purchase bill and the other about drivers waiting at low charge or leaving short of what they booked, and I am still waiting for direction from the regional office before deciding which of the two should rank higher. Until then I am not giving a ranking between them. Please do not assume one; both matter, but their relative importance is open for now.

r16 (Negation or concession, writer B, expected: voltage > service > cost)

To be clear about tomorrow's priorities: this is not a cost-saving exercise. Paying somewhat more for electricity is acceptable if that is what it takes. Nor is driver service the thing we put before everything else. What must be avoided above all is a schedule that pushes bus voltages outside their limits; protecting voltage is the first thing the dispatch will be judged on. That said, drivers are not an afterthought. Of the remaining concerns, service matters more than the electricity bill, so cars should not sit at a low battery for long, and they should not leave without the charge they requested, even if keeping that promise costs us a bit extra. Cost comes last, though I would still rather not waste money needlessly.

r17 (Multi-party ruling, writer A, expected: cost > service > voltage)

Three views came in on how to evaluate the day. Grid operations said keeping bus voltages within limits should dominate. The charging-network team argued for putting drivers first: short waits when batteries are low and full delivery of requested charge. Procurement asked that electricity purchase cost lead, since the budget is under pressure. I have heard everyone, and as the shift supervisor I am making the call. Procurement's position wins: purchase cost is the top priority. The charging-network team's concern comes next, ahead of grid operations' request. Voltage limits remain part of the evaluation but matter least of the three. Please apply this ruling, not any individual team's wish.

r18 (Superseded instruction, writer B, expected: voltage > service > cost)

Update to the dispatch guidance. On Monday I circulated a note asking you to put purchase cost ahead of everything else, with drivers second and the network last. Please disregard that note entirely; it was written before the substation team reported how weak the end of the feeder is under evening load, and it no longer applies. The current guidance for judging a full day of charging is this. Voltage security is the primary concern, so the schedule should keep bus voltages within their limits. Next comes the service we give to EV owners: short waits when a car arrives nearly empty, and full delivery of what each driver requested before departure. Electricity cost is now the lowest of the three priorities. I still want it watched, but only after the first two are taken care of.

r19 (Negation or concession, writer B, expected: cost > service > voltage)

I want to head off a misunderstanding about how tomorrow will be judged. It is not voltage first this time. The feeder can tolerate a few modest excursions, and we are willing to accept a somewhat weaker voltage profile if that is the price of the other goals. It is also not service above all else. The main measure is how much we pay to buy electricity across the day; keeping that down comes first. Customer service is not being dropped either: between service and voltage, service is clearly the more important, so avoid leaving vehicles at low charge for long and avoid sending them away short of their requested energy. Voltage comes after both. It is still worth keeping reasonable, but it gives way when it conflicts with savings or service.

r20 (Multi-paragraph handover, writer A, expected: voltage > service > cost)

Handover note, day desk.

Overnight was quiet. The night team logged a brief alarm on one of the far-end buses, which cleared without action; no equipment faults remain open.

Station staff mentioned that a few drivers complained about sitting at very low charge for a long time, and one left without the energy she had booked. Customer relations would like this kept in view.

For today's evaluation, the standing task is unchanged from what the planning group set: voltage security on the feeder is the leading criterion. Charging service for the vehicles is next, ahead of electricity purchase cost, which is the least of the three.

Finance asked whether cost could be moved up; that request was not approved, so keep the order above.

r21 (Cross-sentence reference, writer C, expected: voltage > service > cost)

Three things get weighed when a day of dispatch is reviewed: the condition of the feeder, meaning whether bus voltages stay within their limits; how much we pay for power; and how well the vehicles are charged, meaning no long waits at low battery and no cars leaving without their requested energy. The first of these is the main concern, and it outranks everything else when you assess a day. The other two are both secondary to it, but they are not equal: the third matters more than the second, so a cheaper purchase bill should not excuse poor charging. In other words, after the feeder comes the item I named last, and the one I named in the middle sits at the bottom.

r22 (Superseded instruction, writer A, expected: cost > service > voltage)

Correction to the standing instruction. Until this morning the desk was told to rank feeder voltage above everything, with service in the middle and cost trailing behind. That instruction has been cancelled by the operations manager and no longer applies. The replacement is as follows. Our main yardstick for the whole day is the cost of the electricity we buy from the upstream grid; a dispatch that keeps that bill down is the better dispatch. Next in importance is the service the chargers give to drivers, meaning they should neither be left waiting at a low state of charge nor leave without the charge they requested. Keeping bus voltages within limits is now the lowest of the three, though it is not something to neglect entirely.

r23 (Background + final directive, writer B, expected: voltage > service > cost)

Some context before tomorrow's plan. The feeder has been carrying heavy evening residential load all week, and with very little solar on the far laterals, the tail-end buses have been sagging every night once commuters plug in. Procurement also mentioned that wholesale prices have crept up, and the charging hub keeps forwarding complaints

from drivers. None of that changes the basic picture, so here is how I want tomorrow's whole-day dispatch judged. Keeping every bus voltage inside its limits comes first. After that, look after the drivers: nobody should sit for a long time on a low battery, and nobody should drive away without the charge they asked for. What we pay for electricity matters least of the three; it still counts, but it should not be bought at the expense of the grid or the drivers.

r24 (Negation or concession, writer A, expected: cost > service > voltage)

This is not a voltage-first day, and nobody should judge it as one. The feeder can live with somewhat weaker voltage margins if that is the price of a cheaper or better-served day; voltage is the last of our three concerns, not an absolute veto. Nor is driver service the headline. What I care about most is not overpaying for the electricity we purchase. If a schedule saves money at the expense of slightly longer low-battery waits or a little less energy delivered before departure, that is a trade I accept. Still, service is not irrelevant: when costs are comparable, the schedule that serves vehicles better wins, and service should not be sacrificed merely to tidy up voltages.

r25 (Multi-paragraph handover, writer B, expected: cost > service > voltage)

Handover note, evening shift.

The substation crew finished their maintenance at four and the feeder is back in its normal configuration. They report healthy voltage margins on all laterals.

From operations: for the coming day, judge the charging schedule mainly on the money spent buying electricity. That is the headline concern from the commercial side and it has been agreed with the shift lead.

The customer desk asked me to pass on that they still care about service, and the shift lead confirmed it ranks right behind cost: drivers should not wait long on a low battery and should not leave short of what they requested.

On voltage, the shift lead's view is that with margins this healthy it is the least pressing of the three. Keep it in view, but it should not override cost or service.

Nothing else outstanding.

r26 (Background + final directive, writer B, expected: cost > service > voltage)

For background: last quarter the finance team flagged that our upstream energy purchases ran well over budget, mostly because charging bunched up during expensive hours. The network side has been quiet. The feeder has had spare headroom on most days, and voltage excursions have been rare and mild. Drivers have generally been satisfied with the hub. With that in mind, the instruction for the coming day is as follows. Judge the dispatch first on how little we spend buying electricity. The second consideration is charging service, meaning vehicles should not be left waiting at a low state of charge for long and should depart with the energy they requested. Voltage security sits at the bottom of the list for this day: keep an eye on it and avoid needless excursions, but do not trade away cost savings or customer service to polish voltage margins.

r27 (Multi-paragraph handover, writer C, expected: voltage > service > cost)

Handover notes, control room.

Equipment: the capacitor bank at the far end of the feeder was inspected this morning and is back in service. No outstanding alarms.

Charging hubs: steady demand all day, nothing unusual. A few chargers were reported slow by drivers; maintenance has a ticket open.

Evaluation task carried over from the supervisor: the dispatch for the whole day should be judged first on voltage security, keeping every bus inside its limits. After that comes the charging service given to vehicles: avoid long waits with low batteries, and make sure they leave with the charge they asked for. Behind both sits the cost of the electricity we buy, which should be kept in mind but does not outrank the other two.

Nothing else pending. The same task applies to the next shift; please confirm on arrival.

r28 (Negation or concession, writer C, expected: voltage > service > cost)

Before anyone scores the day, a word on what this evaluation is not about. Cost is not the first thing to optimise, and squeezing every last bit of convenience out of the charging sessions is not the main goal either. A higher electricity bill is a price I am willing to pay to keep the network healthy, and drivers can tolerate a slightly less ideal session for the same reason. The first thing I will judge is whether the dispatch kept bus voltages within their limits; nothing ranks above that. Between the remaining two, the drivers come before the bill: long waits on a low battery, or vehicles leaving short of their requested charge, matter more than a higher purchase cost. Cost is still tracked, just at the bottom.

r29 (Cross-sentence reference, writer A, expected: voltage > service > cost)

Two things keep coming up when we review dispatch: what we spend on electricity, and how well drivers are served at the chargers, which covers both avoiding long waits at low charge and making sure they leave with the charge they requested. Between these two, the latter matters more to us than the former. There is a third concern that outranks both of them, and that is voltage security on the feeder, keeping every bus within its limits. This point should guide the evaluation above all else. So when you judge the day, start from this point, then look at the service matter I described, and only after that consider the spending.

r30 (Multi-party ruling, writer B, expected: cost > service > voltage)

Three views were put forward at the planning meeting. Grid operations asked for voltage to be the overriding priority. The hub manager asked that customers be protected above everything else. The commercial team wanted the day judged mainly on what we spend on electricity. The regional director, who has the final say, closed the discussion with the following decision. For the whole of tomorrow, purchase cost is the top criterion. The hub manager's point comes next: service to EV drivers, meaning short waits at low charge and departures with the requested energy, ranks second. Grid operations' request is not adopted as the lead priority; voltage security ranks third. It still counts and the network should not be neglected, but within this evaluation it is the least important of the three.

r31 (Multi-paragraph handover, writer C, expected: cost > service > voltage)

Shift handover, overnight crew to day crew.

Weather was mild and the feeder was lightly loaded. Voltage readings stayed comfortably within range; the monitoring team has no concerns to pass on.

Finance stopped by and reiterated what the operations director decided last week, which remains in force: when we judge a day's charging dispatch, the electricity purchase cost is what counts most. Driver service comes after cost, so keep waiting times short for vehicles that arrive nearly empty and make sure cars leave with the charge they requested. Voltage limits are the remaining item, still assessed but ranked behind both cost and service.

The car park barrier is sticking again; facilities have been told.

That's everything. Carry the same evaluation criteria through the full day.

r32 (Missing priority (should clarify), writer C, expected: clarify)

Quick note on how we'll be judging dispatch. Several people raised the charging experience, and I agree it should count for more than voltage: a day where vehicles wait long at low charge or leave short of their requested energy is worse than one where bus voltages run closer to their limits. Voltage is still part of the review, of course. The electricity bill will also be looked at, as it always is. Where it fits relative to the other two hasn't been decided; procurement and the network team are still arguing it out and there's no ruling yet. Apply this to the whole day.

r33 (Missing priority (should clarify), writer B, expected: clarify)

A quick note on tomorrow's charging dispatch. The feeder has been under some strain in the evenings, so I want voltage security to be treated as more important than the electricity bill when the two conflict. Bus voltages should stay inside their limits. Please also remember the drivers: we have had a few complaints about cars sitting at a low battery for a long time and about vehicles leaving with less charge than they asked for. Service matters to us and should be part of how the day is assessed. Use your judgement on the rest and send me the schedule once it is ready.

r34 (Superseded instruction, writer C, expected: cost > service > voltage)

Update to the dispatch evaluation criteria. Until today we had been judging charging schedules primarily on the experience of the drivers, with voltage limits as the next concern and the electricity bill treated as an afterthought. That arrangement is cancelled as of this message; please don't carry any of it forward. The new criteria are these. The cost of purchasing electricity is now what matters most when comparing one day's dispatch with another. Service to the vehicles, meaning short waits when batteries are low and departures with the requested charge, sits in the middle. Keeping bus voltages inside their limits is the lowest of the three. It still belongs in the evaluation and we are not dropping it, but it yields to both of the others.

r35 (Missing priority (should clarify), writer C, expected: clarify)

For today's dispatch evaluation, the main thing is the feeder. Bus voltages need to stay within their limits, and any schedule that lets them stray should be judged harshly, ahead of anything else we look at. Beyond that, I'd like the assessment to also reflect how the drivers were served, meaning whether vehicles waited long at a low state of charge and whether they left with what they asked for, and what we ended up paying for electricity. Both of those belong in the evaluation and both should be taken seriously. I haven't settled in my own mind which of them deserves more weight, and I'd rather not guess, so I'm leaving that open for now. Please use this for the whole day.

r36 (Cross-sentence reference, writer B, expected: voltage > service > cost)

Two things tend to pull against each other on this feeder in the evening: keeping bus voltages within their limits, and keeping the electricity bill down. For tomorrow, the former clearly outranks the latter. There is a third concern as well, namely the service drivers get at the charging points: not waiting a long time with a nearly empty battery, and leaving with the charge they asked for. This point sits between the two I named first. It yields to the one that outranks, and it takes precedence over the one that is outranked. So whenever the schedule has to give something up, the bill is what gives first, and the network is what gives last.

r37 (Superseded instruction, writer B, expected: cost > service > voltage)

Please note a change of instructions. The guidance issued at the start of the month, which put customer service at the top, voltage second and price last, has been withdrawn by the operations manager and should not be used for any further evaluation. That arrangement made sense during the launch promotion, which is now over. From today, a day of charging dispatch is to be assessed with electricity purchase cost as the leading criterion. Charging service follows in second place: we still do not want vehicles stuck at low charge for long periods, and we still want them to leave with what they asked for. Holding bus voltages within limits is the third and least important concern under the new guidance. It is not to be ignored, but it yields to the other two.

r38 (Multi-paragraph handover, writer A, expected: cost > service > voltage)

Shift handover, dispatch evaluation.

Network status: no outages, voltage readings have been within the normal range for most of the week, and maintenance has nothing scheduled.

Customers: the charging desk passed on some feedback about cars waiting at low battery and departing without their full requested charge. They want service treated as an important item, not a forgotten one.

Current task: management confirmed this morning that electricity purchase cost is the main criterion for judging the whole day. Charging service comes after cost but before voltage. Voltage security is the lowest of the three priorities, though it is still included.

Reminder: an old sticky note on the console still says 'voltage first'. It predates this instruction and should be ignored.

r39 (Multi-party ruling, writer A, expected: voltage > service > cost)

We met this morning to agree how today's dispatch should be judged. The finance representative argued that purchase cost should come first, given the recent price increases. The customer-relations lead wanted charging service at the top, pointing to complaints about cars stuck at low charge and drivers leaving without the energy

they booked. The network engineer insisted that voltage problems on the feeder are the real risk. As duty manager, the decision is mine, and here it is: voltage security leads. Customer service comes second; I take the complaints seriously, and they outrank the bill. Purchase cost comes third, and finance's concern is noted but yields to the other two. That is the ruling for the whole day.

r40 (Multi-party ruling, writer C, expected: voltage > service > cost)

We had a long discussion about how to score a day of dispatch. Priya from procurement argued that purchase cost should lead, since that is what the monthly review looks at. Tom from customer operations wanted driver service at the top: short waits for low-battery vehicles and full delivery of the requested charge. Lena from network planning pointed out that the feeder's voltage margins are thin and should not be traded away. As the manager responsible for this evaluation, here is my decision, and it is the one that applies. Voltage security on the feeder comes first. Tom's concern about driver service comes next, and I agree it is important. Priya's point about cost is taken on board, but cost ranks last of the three. Please judge every full day of dispatch on that basis.

r41 (Missing priority (should clarify), writer A, expected: clarify)

At the coordination meeting, the network engineer argued that keeping bus voltages within limits must lead the whole-day evaluation, with driver charging service next and purchase cost last. The procurement lead disagreed sharply and wanted electricity purchase cost at the top, with service second and voltage last. The customer team took no side on the top item, only saying that drivers should not wait long at low charge or leave short of their requested energy. The meeting ended without a decision; the duty manager was absent, and nobody else present has the authority to choose between the two positions. The minutes record both views and note that a ruling is still pending.

r42 (Background + final directive, writer A, expected: voltage > service > cost)

Some context before the instruction. Last month the feeder saw several evenings where residential demand climbed while rooftop solar had already faded, and the far end of the lateral sagged noticeably. Our protection engineers flagged it, and the regional office has asked every dispatch desk to take it seriously. At the same time, drivers keep telling the station staff that they hate sitting at a nearly empty battery or driving away with less charge than they asked for, and the finance team has been watching our purchase bills closely. Here is how I want today's dispatch judged. Keeping bus voltages inside their limits is the thing that matters most. After that, look at how well the vehicles are served: short waits at low charge, and leaving with what they requested. Electricity purchase cost is the least important of the three, though it should not be thrown away for nothing.

r43 (Contradiction (should clarify), writer A, expected: clarify)

Here is the current ranking for today's whole-day evaluation. Voltage security on the feeder leads, and nothing should be placed above it. Charging service for drivers, meaning short waits at low charge and leaving with the requested energy, ranks second, and the electricity purchase bill comes third. However, please also treat the purchase bill as the top concern of the day, ahead of voltage, because finance has made that the leading criterion. Both statements belong to the same current instruction; neither replaces the other, and I expect the dispatch to be judged by both at once.

r44 (Contradiction (should clarify), writer C, expected: clarify)

On judging the charging dispatch: the drivers' experience outranks the electricity bill. Long waits with low batteries and cars departing short of their requested charge are worse than paying more for power. Voltage within limits sits below both of those. That is the standard for the whole day. Also, to be clear, for the same day and the same evaluation, keeping the bus voltages within limits should be the most important factor, and purchase cost should rank ahead of how the vehicles are served. I'm not withdrawing either part; treat everything in this note as current.

r45 (Contradiction (should clarify), writer B, expected: clarify)

Instructions for tomorrow's whole-day dispatch. The commercial position this month is tight, so the schedule should be judged first and foremost on how little we spend buying electricity; nothing else should come ahead of that. Driver service is the second consideration: no long waits at low charge, and cars should leave with the energy they

asked for. One more thing from the same review: voltage security must be the top priority for the day, ahead of cost and ahead of service, because the end of the feeder has been running close to its limits. Service still sits in the middle. Please build the plan on this basis and do not come back to me with questions.

r46 (Contradiction (should clarify), writer B, expected: clarify)

Handover note, day shift.

The operations manager confirmed this morning that, for the coming day, charging service is to be ranked above electricity cost: drivers should not wait long on a low battery and should leave with their requested energy, even if that means paying more.

The same manager also confirmed that bus voltages within limits remain the leading concern for the whole day.

In the afternoon call, the manager added that the electricity bill must take precedence over driver service tomorrow, since the budget review is still open. Nothing said in the morning was withdrawn, and both instructions are to stand as given.

No equipment faults to report.

r47 (Multi-party ruling, writer B, expected: voltage > service > cost)

Minutes from this morning's dispatch call. The finance lead argued that tomorrow should be run to minimize energy purchase cost, since prices are expected to be high. The customer liaison wanted driver service put first, pointing to recent complaints about cars left nearly empty for hours. The protection engineer warned that the end of the feeder is running close to its lower voltage limit and asked that voltage be the top concern. After hearing everyone, the duty manager made the call, and that decision is what the dispatch will be evaluated against. Voltage security leads: keeping bus voltages within limits is the first priority. Service to drivers comes second, covering both waiting at low charge and leaving short of the requested amount. Purchase cost comes third; finance's concern is noted, but it ranks below the other two.

r48 (Negation or concession, writer C, expected: cost > service > voltage)

This is not a grid-security study, and voltage is not the first thing we're judging. Nor is the goal to give every driver the best possible session regardless of what it costs. The day's dispatch should be assessed above all on how much we spend buying electricity. Slightly longer waits, or the occasional shortfall against a requested charge, are an acceptable price for a cheaper day, though service still has to be weighed and it outranks the voltage question. Some movement of bus voltages toward their limits is tolerable too; voltage still counts, only less than service and cost. Put simply, don't let concern for voltage override the drivers, and don't let concern for the drivers override the cost of energy.

r49 (Background + final directive, writer C, expected: cost > service > voltage)

For context: the utility renegotiated our supply contract last month, and finance has flagged that the depot's electricity bill is now the line item under the most scrutiny. The grid side has been quiet; the voltage profile on this feeder has had plenty of margin all season, and drivers have mostly been satisfied with the service. With that said, this is how I want each day of charging dispatch evaluated from now on. The purchase cost of electricity is the primary yardstick. Next in importance is charging service: vehicles shouldn't sit for long at a low state of charge, and they shouldn't leave without the energy they requested. Keeping bus voltages within limits remains part of the assessment, but it ranks behind both cost and service.

r50 (Missing priority (should clarify), writer B, expected: clarify)

Guidance for the coming day. Our first priority is the people who use the hub. A vehicle that arrives nearly empty should not be left waiting for long, and every driver should leave with the charge they requested. That is what the day will be judged on before anything else. Beyond that, there are two further things in play: keeping bus voltages inside their limits on the feeder, and what we pay to buy electricity. Both of them matter and neither should be ignored. I am not going to rank them against each other here; just make sure the schedule takes them into account after the drivers are looked after.

r51 (Standard wording, writer A, expected: cost > service > voltage)

Electricity purchase cost comes first, EV charging service second, and voltage security last.

r52 (Background + final directive, writer C, expected: voltage > service > cost)

A bit of background first, then the actual instruction. Over the past few weeks the feeder has been running close to its limits on several occasions, mostly on the long laterals far from the substation, and the protection team has asked us to stop treating that as a minor nuisance. Meanwhile the charging hubs have been fairly busy, and the purchasing desk keeps reminding everyone that wholesale prices have been volatile. None of that background is the instruction itself, so here it is plainly. When you judge a full day of charging dispatch, the first thing to look at is whether bus voltages stayed inside their limits. After that, judge how well the drivers were served: vehicles should not be kept waiting long while their batteries are low, and each one should depart with the energy its driver requested. The electricity purchase bill comes last; it still counts, but only after the other two.

S7. Fig. S1: battery-degradation retraining

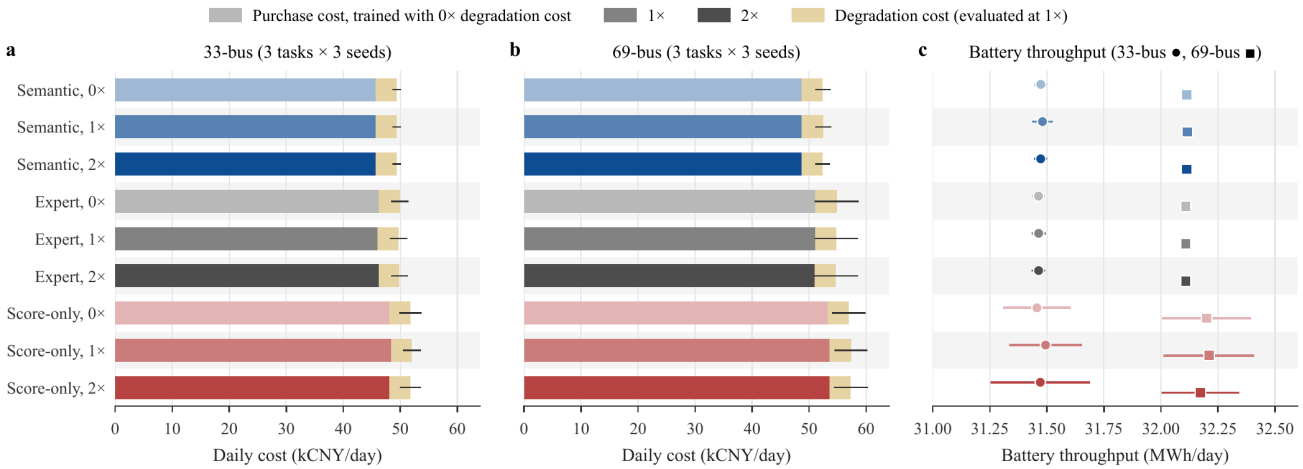


Fig. S1. Purchase and battery-degradation costs on the 33-bus (a) and 69-bus (b) systems and battery-side throughput on both systems (c) after retraining with zero, one, and two times the nominal degradation coefficient ($c_{\mathrm{deg}} = 116.6$ CNY/MWh); means over three preference settings and three seeds. Degradation costs are evaluated at the nominal coefficient. The corresponding operating metrics are given in Table V of the manuscript; all rows, including the zero coefficient, come from the retraining runs of the degradation study.

S8. Table S1: daily network-loss energy

Table S1. Daily network-loss energy (MWh/day) over the 24 held-out days for the policies frozen after 10,000 episodes; means and standard deviations across three seeds. The policies charge almost the same daily EV energy (32.9 to 33.2 MWh on the 33-bus and 33.8 to 34.1 MWh on the 69-bus system), so a daily loss ratio normalized by native load plus EV charging ranks the policies in the same order as the loss energy. For these three methods, the seed means of this table are the network-loss row of Table III of the manuscript.

System, preference	Score-only Reward	Semantic Reward	Expert Reference
33-bus, voltage first	10.44 ± 2.31	6.64 ± 0.29	9.75 ± 0.70
33-bus, service first	9.22 ± 4.23	6.00 ± 0.26	5.55 ± 1.08
33-bus, cost first	9.22 ± 2.55	4.26 ± 0.01	4.32 ± 0.00
69-bus, voltage first	17.32 ± 7.41	10.08 ± 0.38	20.44 ± 0.86
69-bus, service first	18.02 ± 5.50	7.75 ± 0.63	10.36 ± 1.58
69-bus, cost first	13.57 ± 2.64	4.94 ± 0.18	5.13 ± 0.01

S9. Experimental settings


```
{
  "protocol_version": "revised manuscript (TSG-02158-2026 revision), 2026-10-29",
  "systems": {
    "33": {
      "feeder": "modified IEEE 33-bus distribution network, 10 EVCSs",
      "observation_dim": 59
    },
    "69": {
      "feeder": "IEEE 69-bus feeder (12.66 kV, 3.80 MW nominal load), 10 EVCSs",
      "observation_dim": 95
    }
  },
  "preferences": {
    "V": {
      "name": "voltage first",
      "objective_order": [
        "voltage security",
        "charging service",
        "cost"
      ],
      "priority_metric": "voltage violation rate"
    },
    "S": {
      "name": "service first",
      "objective_order": [
        "charging service",
        "voltage security",
        "cost"
      ],
      "priority_metric": "waiting deficit"
    },
    "C": {
      "name": "cost first",
      "objective_order": [
        "cost",
        "charging service",
        "voltage security"
      ],
      "priority_metric": "purchase cost"
    }
  },
  "policy_training": {
    "total_episodes": 10000,
    "seeds": [
      503,
      509,
      521
    ],
    "seed_role": "policy and reward-model initialization and training-day sampling; annotations
    ↪  are generated from each run's trajectories",
    "development_days": 8,
    "held_out_evaluation_days": 24,
    "evaluation_actions": "deterministic mean actions",
    "outer_iterations_K": 100,
    "episodes_per_iteration": 100,
```

```
  "labeled_steps_per_run": "2400 (100 operating days x 24 steps; 60/20/20 days for fitting,
  ↪  early stopping, held-out evaluation)",
  "service_constraint": {
    "scope": "all methods, including the PSA baseline",
    "form": "end-of-day penalty sum_m mu_m * max(0, K_m - limit_m), K_m in {waiting deficit,
    ↪  departure shortfall}",
    "multiplier_update": "dual ascent after every block of 20 episodes",
    "multiplier_cap": 5,
    "service_limits_met": "share of held-out days on which both K_m are within their limits
    ↪  (Table III row)"
  }
},
"reward_model": {
  "architecture": "MLP with shared encoder, scalar head and semantic projection head",
  "text_encoder": "bert-base-uncased (mean pooling, 768-d), frozen",
  "alignment_weight": 0.5,
  "optimizer": "Adam, lr 1e-3, weight decay 1e-4, batch 64, early-stopping patience 100",
  "split": {
    "mode": "by operating day (all steps of a day in one subset)",
    "fit_days": 60,
    "early_stopping_days": 20,
    "held_out_days": 20,
    "reduced_label_variants_fit_days": [
      10,
      20
    ],
    "same_split_for_score_only_and_semantic": true,
    "results": "rm_held_out_results.md"
  }
},
"annotator": {
  "model": "qwen-turbo",
  "temperature": 0.3,
  "response_format": "json_object",
  "request_timeout_s": 180,
  "max_retries": 10,
  "episodes_per_batch": 2,
  "contrastive_pairs_per_batch": 4,
  "prompt_template": "prompt_template_v6.md",
  "validation_rules": "validation_stages.md"
},
"annotator_comparison_fig11": {
  "seeds": 1,
  "training_episodes": 1500,
  "note": "separate single-seed experiment with its own return scale; compared only within
  ↪  Fig. 11"
},
"psa_baseline": {
  "name": "Price-Service Adaptive Reward (PSA)",
  "step_reward": "-(purchase-cost term + 1.0 * voltage-violation term) / s_sys",
  "s_sys": {
    "33": 5.09,
    "69": 4.95
  },
  "s_sys_definition": "pooled within-profile standard deviation of the PSA daily score on the
  ↪  initialization data",
```

```
  "service": "only through the shared service constraint above",
  "information": "no LLM, operator evaluation, or preference information; identical training signal under V, S, C",
  "other_settings": "same initialization, PPO settings, training-day pool, seeds, and 24 held-out days as the other policies"
},
"voltage": {
  "evaluation_band_pu": [
    0.9,
    1.1
  ],
  "voltage_violation_rate": "fraction of hours in which at least one bus leaves the band",
  "figure13_reference_lines_pu": [
    0.9,
    1.1
  ],
  "figure13_reference_lines_meaning": "voltage safety limits used for the reported voltage metrics",
  "figure13_run": "Semantic Reward, 33-bus voltage-first, seed 503, 10000 episodes; test day 3, reset identifier 2266002",
  "figure13_selection": "Among the violation-free days for the fixed policy, choose the day whose return is closest to the median return across all 24 test days."
},
"battery_degradation": {
  "c_deg_nominal_cny_per_mwh": 116.6,
  "basis": "LFP pack price 81 USD/kWh, 7.20 CNY/USD, 2,500 equivalent full cycles to 80% capacity, each cycle = charge-plus-discharge throughput of twice the capacity",
  "main_comparison_c_deg": 0,
  "sensitivity_multipliers": [
    0,
    1,
    2
  ],
  "sensitivity_evaluation": "purchase-plus-degradation cost at the nominal coefficient; 33-bus and 69-bus, 10,000 episodes, 24 held-out days"
},
"operating_conditions": {
  "conditions": [
    "EV fleet 0.8x",
    "EV fleet 1.2x",
    "low load with high PV",
    "concentrated arrivals"
  ],
  "modes": [
    "retraining under each condition (Table VII, Fig. 9)"
  ]
},
"language_interface_study": {
  "texts": 52,
  "unambiguous": 40,
  "defective": 12,
  "expression_types": 9,
  "writers": "three independent LLM-driven writer agents",
  "models": {
    "initial_run": "qwen-turbo",
    "repeat": "qwen3.8-flash"
```

```
    },
    "temperature": 0,
    "requests_per_text": 3,
    "parser": "keyword parser (v0) and keyword parser with clarification fallback (v1)",
    "texts_and_outcomes": "language_interface_texts.md"
  },
  "cost_display": "kCNY/day"
}
```